

A SHORT READ BEFORE OUR CLASS

Python: Operators & Data Types



Let's explore how a snack shop can use Python.

You do not need to memorize anything. Just get familiar with the ideas.

How to use this pre-read

This is a first look—not a test.

1 • Read

Go through the slides at your own pace.

2 • Notice

Look at the examples and think what they mean.

3 • Bring

Bring one question or one thing you noticed to class.

It is completely okay if some ideas feel new. We will practise together.

Our example: a small snack shop

Imagine you are helping a shop manager.



CHIPS

Price

\$2

Packets

5

Available?

Yes

Python can store these details and help the manager make decisions.

A variable is a labelled box

Python uses a name to remember a value.

Example code

```
snack_name = "Chips"  
price = 2  
quantity = 5
```

Read it like this

snack_name stores the text Chips.
price stores 2.
quantity stores 5.

Think of a variable as a box with a name. Python uses that name to remember the value stored inside the box.

Values can be different types

Text (str)

"Chips"

A word or text

Whole number (int)

5

A number without
a decimal

Decimal (float)

2.50

A number with a
decimal

Boolean (bool)

True / False

Yes or no

Python uses the type to understand what kind of value it is working with.

Operators help Python do a job

An operator is a symbol that tells Python what to do.

Add: +

$$2 + 3 = 5$$

Use + to add numbers.

Multiply: *

$$\$2 \times 5 \text{ packets} = \$10$$

In Python, use * for multiply.

Shop question: If one packet costs \$2 and you buy 5, what is the total?

Comparisons ask a yes/no question

Less than: <

$2 < 5$
True (yes)

Greater than: >

$7 > 5$
True (yes)

Equal to: ==

$2 == 2$
True (yes)

A comparison gives an answer of True or False.

A condition can lead to an action

Python example

```
quantity = 2  
if quantity < 5:  
    print('Restock chips')
```

What does it mean?

Is quantity less than 5?
Yes → True
So Python displays:
Restock chips

The shop manager can use a condition to decide what to do next.

Python can also work with text

Join text with +

```
shop = 'Quick ' + 'Bites'
```

Result: Quick Bites

Count text with len()

```
name = 'Chips'  
len(name)
```

Result: 5

Text is called a string. We can join it or count its characters.

Indexing: choose one character

Indexing lets us get one character from a piece of text.

Example code

```
name = 'Chips'  
print(name[0])
```

What does [0] mean?

The number inside [] is the index.
Index 0 means the first character.

Each character has a position (index):



name[0] gives C • name[1] gives h • name[4] gives s

Think about these before class

You can answer in your head or talk it through with someone.

Question 1

A snack costs \$3. You buy 2.
Write python code to find total?

Question 2

There are 2 packets left.
Is $2 < 5$ True or False?
Write python code to print whether restock
is need based on the number of packets left

We will discuss the answers during class.

Swapping: switch two values

We may explore this idea together during class.

Example

```
a = 2
b = 5
temp = a
a = b
b = temp
```

Follow the values

temp keeps 2 safe
a becomes 5
b becomes 2

Think of temp as a temporary holding box. No need to learn this yet.